Natural frequencies and transient impact response of a 3D-printed carbon-fibre PETG fan housing, with an unmeasured elastic modulus

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Abstract

The housing of a desk fan printed in carbon-fibre PETG (PETG-CF) is analysed with code_aster for resonance with the rotor and for its response to an impact. The free-free modes, with the motor and electronics as distributed masses, were first computed with linear tetrahedra; a convergence study shows these to be 12 to 20 % too stiff, while quadratic tetrahedra on three meshes agree within 0.5 %, at 44.0, 64.3 and 109.9 Hz for the first three modes with $E = 6$ GPa. The Young's modulus of the printed material is not measured. Over the published range of 3.8 to 6.0 GPa, modes 3, 4 and 5 cross the third and fourth rotor harmonics and the blade-passing frequency at 2 000 rpm, so a resonance cannot be ruled out. Three impact cases, a 251 g, 2.5 ms pulse with the contact patch held fixed, are run to 30 ms with linear transient dynamics and Rayleigh damping; verification found that the original oblique case fixed a slice through the housing, and it was replaced. With quadratic elements and 5 % damping the peak von Mises stresses away from the contact, 114 to 174 MPa, sit at junctions of the support arms and the stand, and exceed the strength reported for printed PETG-CF, while the 95th percentile stays below 23 MPa.

1 Introduction

Small rotating machines printed in fibre-reinforced polymers, such as desk fans, drone frames and robot covers, need two quick structural checks before a first print: whether a natural frequency of the printed part sits on an excitation line of the rotor, and where the part concentrates stress when it is knocked over. Both are routine with finite elements, but for a printed composite the inputs are uncertain. Fused-filament parts are anisotropic and weaker between layers [1], and short carbon fibres raise the stiffness by an amount that depends on fibre content and print orientation [2–4].

This study covers the housing of a desk fan designed at Polytechnia: a cylindrical shroud on a stand, printed in carbon-fibre PETG (PETG-CF), carrying a brushless motor in the shroud and the electronics in the base. The rotor has five blades and runs at a nominal speed of 2 000 rpm in this study. The contributions are:

- a free-free modal model of the housing with the motor and electronics as distributed masses, checked against the rotor harmonics, with the unmeasured Young's modulus carried as an uncertainty band rather than a single value;

*University of California, Berkeley (MEng). Work carried out in June 2026 at Polytechnia, an engineering consultancy co-founded by the author; convergence study and extended impact runs in October 2026. Repository: `https://github.com/clementdumeril/ventilator-modal-analysis-petgcf`.

- a mesh and element-order convergence study of the first modes, missing from the original analysis;
- a linear transient analysis of three impact cases, with damping, run past the end of the load pulse to the peak response, with stress metrics that separate the boundary-condition singularity at the contact from the stress in the housing, and a check of the contact patches that corrects one of the original cases.

2 Related work

The anisotropy of fused-deposition parts was characterised early by Ahn et al. [1]: strength and stiffness depend on raster orientation and drop across layers. Carbon-fibre reinforcement of printed polymers raises the in-plane stiffness when fibres align with the extrusion direction [2]. For PETG-CF printed by fused filament fabrication, Prabhakaran et al. [3] report a tensile modulus rising from 2.3 GPa for pure PETG to 3.8 GPa at 30 % fibre, with natural frequencies of test beams 18 % to 25 % higher; Batista et al. [4] and Kohn et al. [5] report that print parameters and build orientation change the mechanical response markedly, and that reinforcement does not always raise strength. The rotor-dynamic check used here is the Campbell diagram: frequencies of the structure plotted against rotor speed together with the lines $k\,\Omega$ of the rotor harmonics [6]; for a stationary housing the structural frequencies do not depend on speed. The mesh is generated with NETGEN [7] in SALOME [8]; modal and transient analyses use code_aster [9], with Newmark's average-acceleration scheme [10] for the transient runs.

3 Model

3.1 Geometry and mesh

The housing geometry comes from the Polytechnia FreeCAD model. The original CAD would not take a conforming tetrahedral mesh because of small faces and intersecting surfaces; it was defeatured (small faces and short edges removed, the detailed support arms replaced by simple ribs, one body removed from a fusion), which lowered the volume by 1.4 %, and oriented with the base on the xy plane and the height along $+z$. The reference mesh (Figure 1) has 95 358 nodes and 312 777 linear tetrahedra (TETRA4), from NETGEN with a minimum size of 1 mm, a maximum size of 10 mm and fineness 2.

3.2 Material and masses

PETG-CF is modelled as an isotropic linear-elastic solid (Table 1). The density is that of the filament. The Young's modulus of 6 GPa used in the original analysis has no documented source; the published value for FFF-printed PETG with 30 % carbon fibre is 3.8 GPa [3]. The results below are computed at 6 GPa and carried over the range 3.8 GPa to 6.0 GPa. The motor (5010 brushless, estimated 0.090 kg) and the electronics (controller board, ESC and switch, estimated 0.075 kg) are not modelled as solids: their mass is added to two spherical groups of elements (radius 30 and 45 mm, elements with at least half of their nodes inside) by raising the density there, without adding stiffness. The total mass is 1.080 kg.

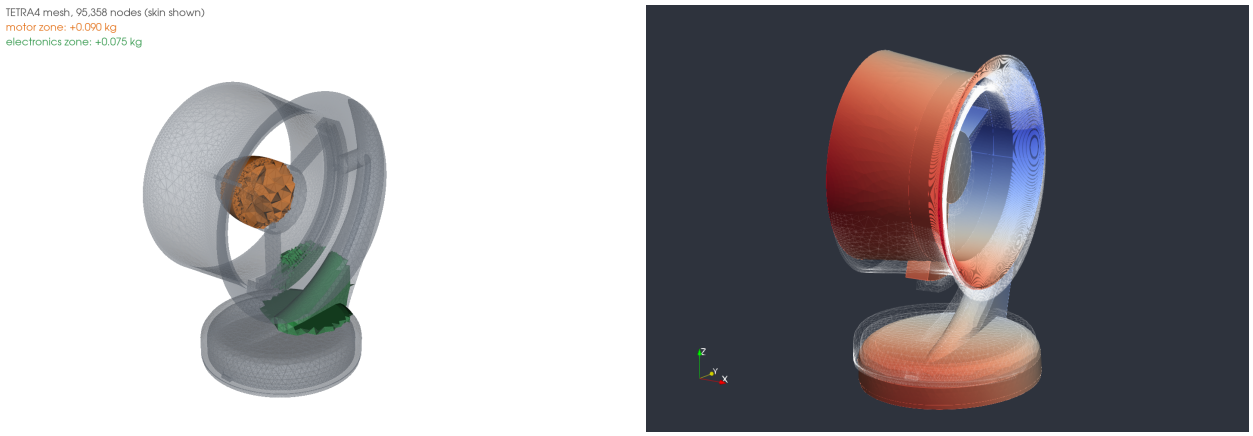


Figure 1: Left: reference mesh (skin shown) with the motor (orange) and electronics (green) mass zones. Right: first elastic mode at $E = 6\,\mathrm{GPa}$ (linear-element run of June 2026; deformation amplified).

Table 1: Model parameters.

Quantity	Value
PETG-CF	isotropic, $E = 6.0\,\mathrm{GPa}$ (range 3.8 GPa to 6.0 GPa), $\nu = 0.35$, $\rho = 1\,300\,\mathrm{kg/m^3}$
Motor zone	sphere $R = 30\,\mathrm{mm}$, 23 127 elements, $+0.090\,\mathrm{kg}$
Electronics zone	sphere $R = 45\,\mathrm{mm}$, 11 616 elements, $+0.075\,\mathrm{kg}$
Total mass	1.080 446 kg
Modal boundary conditions	free-free
Rotor	5 blades, nominal 2 000 rpm ($n = 33.3\,\mathrm{Hz}$, BPF $= 166.7\,\mathrm{Hz}$)
Impact pulse	triangular, 2.524 ms, peak $2.466 \times 10^4\,\mathrm{m/s^2}$ (251 g), $\Delta v = 3.11\,\mathrm{m/s}$

3.3 Modal problem

The free-free modes solve $\mathbf{K}\boldsymbol{\phi} = \omega^2\mathbf{M}\boldsymbol{\phi}$. The six rigid-body modes lie below 1 Hz; the search band starts at 1 Hz. For an isotropic material at fixed density and Poisson ratio, $\mathbf{K}$ is proportional to E and the mode shapes do not change, so $f(E) = f(E_0)\sqrt{E/E_0}$ exactly. Each mode is compared with the rotor lines $k\,n$ ($k = 1 \ldots 4$) and with the blade-passing frequency $5n$.

3.4 Impact model

The impact is idealised: the nodes of a small patch where the housing touches the floor are held fixed, and the rest of the housing receives a uniform acceleration pulse $a(t)$ (triangular, 2.524 ms, peak 251 g). The patch is the set of nodes within a tolerance of the floor plane, which is tangent to the housing at its extreme point in the impact direction. In the frame of the patch this is

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = -\mathbf{M}\,\mathbf{e}\,a(t), \tag{1}$$

with $\mathbf{e}$ the impact direction. The velocity change of the pulse is 3.11 m/s, the impact speed of a free fall of about 0.5 m with no rebound. There is no contact law, friction or rebound. Three directions are studied (Figure 5): lateral on the $-y$ side, where the shroud touches the floor (63 fixed nodes, of the linear mesh, tolerance 3 mm); lateral on the $+y$ side, where the stand touches first (63 nodes, 5 mm); and oblique, at 45° between $-y$ and $-z$, where the edge of the base touches first (107 nodes, 3 mm). The June study defined the oblique case differently; Section 5 explains why it was replaced.

4 Numerical method

Modes are computed with `CALC_MODES` in a frequency band (shift-invert Lanczos with a Sturm count check), code_aster 16.9 for the reference run and 17.4 for the convergence study. The impact runs of June 2026 used `DYNA_NON_LINE` with an elastic constitutive law up to 2.5 ms. Because the problem is linear, the October runs use `DYNA_VIBRA` in the physical basis with the same Newmark average-acceleration scheme ($\beta = 1/4$, $\gamma = 1/2$), $\Delta t = 0.1$ ms, up to 30 ms; the system is factorised once, which brings the cost from about 1 000 s to 66 s of CPU for the first 2.5 ms. Stresses are computed at the archived instants (every 0.2 ms). Because the fixed patch creates a stress singularity, the stress is reported as the maximum, 99th and 95th percentile of the element von Mises stress over the elements whose centroid lies more than 8 mm from the contact plane, next to the raw maximum.

5 Verification

Model checks. The chain was first run with steel properties: scaling the steel first mode (125.7 Hz) by $\sqrt{E/\rho}$ predicts 52 Hz for bare PETG-CF, against 49.2 Hz once the 0.165 kg of components is added. The mesh volume differs from the prepared CAD by 0.24 %. Recounting the mass-zone elements from the mesh file gives the same 23 127 and 11 616 elements as code_aster. The Sturm count finds 21 eigenvalues in 1 Hz to 1 500 Hz and 21 modes were computed, with residual norms between 2×10^{-11} and 4×10^{-8}. Rotating the model between orientations leaves mode 1 at 49.214 Hz.

Mesh and element order. The June analysis used a single mesh of linear tetrahedra. The convergence study (Table 2, Figure 2) remeshes the oriented geometry with NETGEN and changes the element order. Reducing the maximum element size from 10 to 5 mm barely changes the mesh (102 417 to 105 816 nodes), because the element size is set by the thin walls and the curvature, not by the maximum size; refinement was obtained through the NETGEN fineness instead (123 454 and 184 692 nodes). With linear elements the frequencies keep falling as the mesh is refined (mode 1 from 49.2 to 46.5 Hz, mode 3 from 131.9 to 122.3 Hz): linear tetrahedra are too stiff in the bending of thin walls. Quadratic elements (TETRA10) on the reference mesh and on the 102 417-node mesh give 43.97/64.29/109.87 Hz and 43.94/64.22/109.79 Hz, within 0.1 % of each other, and a third, coarser mesh (92 381 nodes, fineness 1) gives 43.88/63.95/109.44 Hz: the three quadratic results agree within 0.5 %. The linear-element values of the June analysis are therefore 12, 14 and 20 % too high for modes 1, 2 and 3, and the quadratic values are used in Section 6.

Table 2: Convergence of the first three elastic modes ($E = 6$ GPa). Nodes: corner nodes of the linear mesh.

Mesh, element	nodes (corner)	mass (kg)	f_1 (Hz)	f_2 (Hz)	f_3 (Hz)
published, TETRA4	95 358	1.0804	49.21	73.18	131.95
h10, TETRA4	102 417	1.0804	49.00	72.66	131.66
fineness 3, TETRA4	123 454	1.0814	47.37	69.80	125.49
fineness 4, TETRA4	184 692	1.0824	46.47	68.48	122.30
coarse (fineness 1), TETRA10	92 381	1.0794	43.88	63.95	109.44
published, TETRA10	95 358	1.0804	43.97	64.29	109.87
h10, TETRA10	102 417	1.0804	43.94	64.22	109.79

Transient solver. The linear solver (`DYNA_VIBRA`) reproduces the June non-linear-solver results at every archived instant up to 2.5 ms (identical to four significant digits: 79.82 MPa, 95th percentile

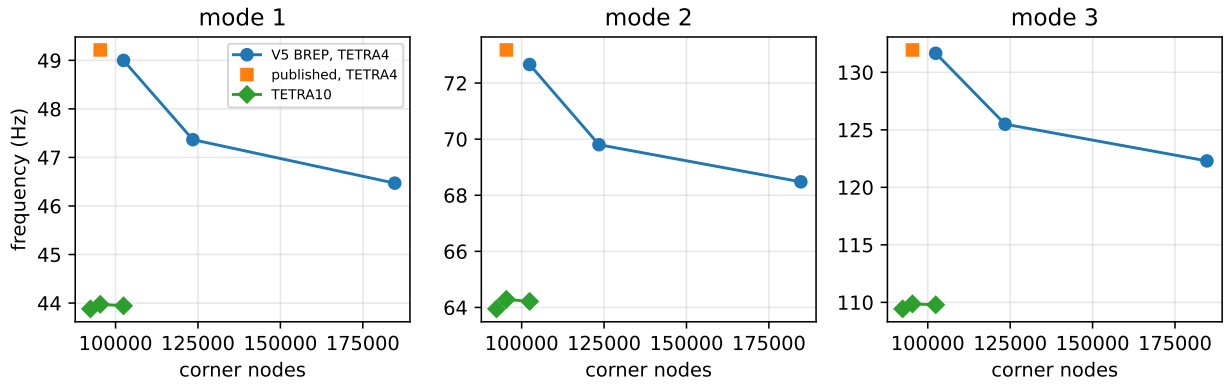


Figure 2: Frequencies of modes 1–3 against mesh size: linear tetrahedra (TETRA4) keep falling with refinement; quadratic tetrahedra (TETRA10) agree within 0.5 % on three meshes.

7.070 MPa, displacement 4.238 mm at 2.5 ms for the $-y$ impact). Halving the time step to 0.05 ms on the same case over 15 ms changes the stress history by at most 0.4 % and leaves the peak unchanged (152.2 MPa at 10.8 ms; displacement 29.23 against 29.26 mm).

Fixed patches. Plotting the fixed node groups on the mesh showed an error in the June oblique case. Its plane, with normal $(0, 1, 1)/\sqrt{2}$, was placed through the $-y$ side of the shroud and is not tangent to the housing: the 1 091 nodes within 1 mm of it form a slice through the shroud, the stand and the base, and 17 295 nodes lie beyond it. That case fixes a cut through the part rather than a contact patch, and its results (peak 91 MPa in June) are not used. A fall in that direction first touches the edge of the base, at the node that minimises $y + z$; the corrected oblique case puts the plane through that node. The planes of the two lateral cases are tangent, with no node beyond them.

Window length. Two cases with quadratic elements and 5 % damping had not settled at 30 ms and were continued to 60 ms: the $-y$ case, whose peak comes at 27.4 ms, and the oblique case, whose response beats. Between 26 and 30 ms both reproduce the 30 ms runs exactly. After 30 ms the $-y$ case reaches 163 MPa at 36.8 ms and at most 121 MPa from 40 to 60 ms, with the displacement decaying to 10.7 mm; the oblique case reaches at most 118 MPa, at 43.8 ms. The 30 ms window therefore contains the damped peaks.

6 Results

6.1 Modes against the rotor lines

With quadratic elements, which the convergence study shows to be converged, the first five elastic modes at $E = 6$ GPa are 44.0, 64.3, 109.9, 151.8 and 192.1 Hz (Figure 3); the published values, computed with linear elements, were 12 to 20 % higher for modes 1–3. In mode 1 the shroud moves on its stand while the base stays nearly still (Figure 1). At 6 GPa no mode sits on a rotor line at 2 000 rpm, but mode 2 is 3.6 % below the second harmonic.

Over 3.8 GPa to 6.0 GPa the modes scale by $\sqrt{E/E_0}$:

- mode 1 spans 35.0 Hz to 44.0 Hz: above the rotation frequency, but only by 5 % at 3.8 GPa;

- mode 2 spans 51.2 Hz to 64.3 Hz, below the second harmonic over the whole range;
- mode 3 spans 87.4 Hz to 109.9 Hz and crosses the third harmonic (100 Hz) at $E = 4.97$ GPa;
- mode 4 spans 120.8 Hz to 151.8 Hz and crosses the fourth harmonic at $E = 4.63$ GPa;
- mode 5 spans 152.8 Hz to 192.1 Hz and crosses the blade-passing frequency at $E = 4.52$ GPa.

Between 4.5 and 5.0 GPa, within the plausible range, three modes cross a rotor line. Whether the printed housing has a resonance at 2 000 rpm therefore depends on a property that was not measured. Mounting adds a further shift: the free-free modes ignore the stiffness of whatever the base stands on. A mass-only estimate of the blades, $f\sqrt{M_0/(M_0 + \Delta M)}$ with $M_0 = 1.080$ kg, lowers every mode by 1 to 6 % for a total blade mass of 25 to 150 g; it ignores where that mass sits.

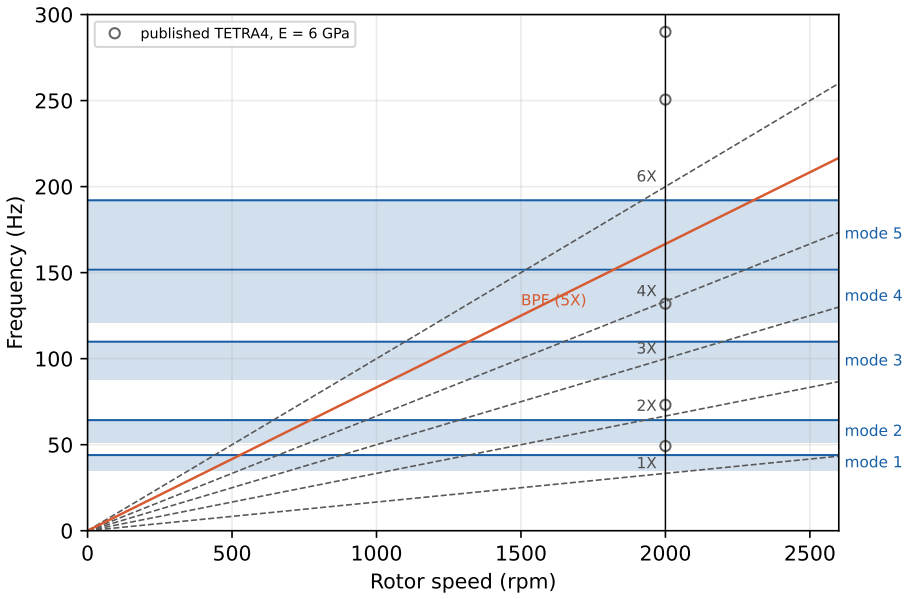


Figure 3: Campbell diagram with the converged (TETRA10) modes. Each band spans the frequency of a mode for $E = 3.8$ GPa (bottom) to 6.0 GPa (top, line); open circles: published TETRA4 values at 6 GPa; dashed lines: rotor harmonics; red line: blade-passing frequency.

6.2 Impact

The June runs stopped at 2.5 ms, at the end of the pulse, while the stress was still rising in the two lateral cases. The October runs continue to 30 ms with linear elements at four damping levels and with quadratic elements undamped and at 5 % (Table 3; the corrected oblique case with quadratic elements at 5 % only). The Rayleigh coefficients give the stated ratio at 15 and 300 Hz and a slightly lower one in between. The damping of printed PETG-CF is not known for this part: Mansour et al. [11] report damping values of 13.8 % for printed PETG and 12.3 % with 20 % carbon fibre from modal tests on specimens, without stating in the abstract whether these are damping ratios or loss factors (a loss factor η corresponds to $\zeta \approx \eta/2$). The range 5 % to 10 % covers both readings; 2 and 5 % are lower, conservative values.

Table 3: Impact peaks over 0 ms to 30 ms: von Mises stress σ (MPa) more than 8 mm from the contact plane, time of the peak t (ms) and height z (mm) of the peak element, for linear (TETRA4) and quadratic (TETRA10) elements and Rayleigh damping ratio ζ. Undamped peaks depend on the length of the window (see text).

		Lateral, Y−			Lateral, Y+			Oblique, base edge		
Element	ζ	σ	t	z	σ	t	z	σ	t	z
TETRA4	0 %	152	10.8	158	123	28.4	182	121	11.8	158
TETRA4	2 %	146	10.8	158	114	2.6	182	117	6.2	158
TETRA4	5 %	138	10.8	158	110	2.6	182	112	6.2	158
TETRA4	10 %	128	10.8	158	104	2.6	182	105	6.2	158
TETRA10	0 %	220	27.4	198	147	11.0	204	–	–	–
TETRA10	5 %	174	27.4	198	114	3.0	180	139	8.2	158

Undamped response. Without damping the housing keeps vibrating after the pulse, and the maximum over the window is set by when the modes happen to add up. With quadratic elements the $-y$ impact shows successive maxima of 166, 169, 179 and 218 MPa at 6, 13, 18 and 27.6 ms, and the displacement is still rising at 30 ms (39.8 mm): the undamped peak grows with the window and is not used as a result.

Damped response. With damping the late maxima of the lateral cases disappear, and the peaks sit at junctions of the housing (Figure 5). For the $+y$ impact, where the stand touches the floor, the peak comes at the end of the pulse (2.6 to 3.0 ms) where the upper support arm meets the motor hub. For the $-y$ impact the housing rocks about the fixed patch (first frequency 16.6 Hz with the patch fixed, linear elements) and the peak comes after the pulse: at 10.8 ms with linear elements, where the stand joins the shroud on the side opposite the impact, and at 27.4 ms with quadratic elements, where the upper arm on the impact side meets the shroud. For the oblique impact on the base edge the peak is again where the stand joins the shroud, at 6.2 ms with linear and 8.2 ms with quadratic elements. With quadratic elements this response beats, with maxima of 134 and 136 MPa between 20 and 30 ms; continued to 60 ms, its later maxima stay lower (at most 118 MPa, at 43.8 ms). Damping lowers the peaks slowly: from 152 to 128 MPa between 0 and 10 % for the $-y$ impact with linear elements. Quadratic elements, which are less stiff in bending, give higher peaks at 5 %: 174 against 138 MPa for $-y$, 114 against 110 MPa for $+y$ and 139 against 112 MPa for the oblique impact. They also move the $-y$ peak to another junction, so the hot spots are best read as a set of regions (arm-to-hub, arm-to-shroud and stand-to-shroud junctions) rather than as single points. The 95th percentile of the stress outside the contact band stays below 26 MPa in all runs (23 MPa with damping), so the peaks are local. The raw maxima, which include the elements at the fixed nodes, reach 310 MPa; inside the contact band they are singularities of the boundary condition.

Comparison with the material strength. The tensile strength of FFF-printed PETG with 30 % carbon fibre is about 60 MPa in the plane of the layers [3]. For unfilled PETG, a manufacturer data sheet gives 38.5 to 44 MPa for flat and on-edge specimens and 19 MPa for upright specimens, which measure the adhesion between layers [12]; Ahn et al. [1] and Kohn et al. [5] report the same weakness across layers. The damped peaks computed here, 114 to 174 MPa with quadratic elements at 5 %, exceed the in-plane strength by a factor of 2 to 3 and the interlayer strength by a factor of 6 to 9. Two readings follow. Taken at face value, the model predicts local failure at the junctions of the arms with the hub and the shroud and of the stand with the shroud. Alternatively the

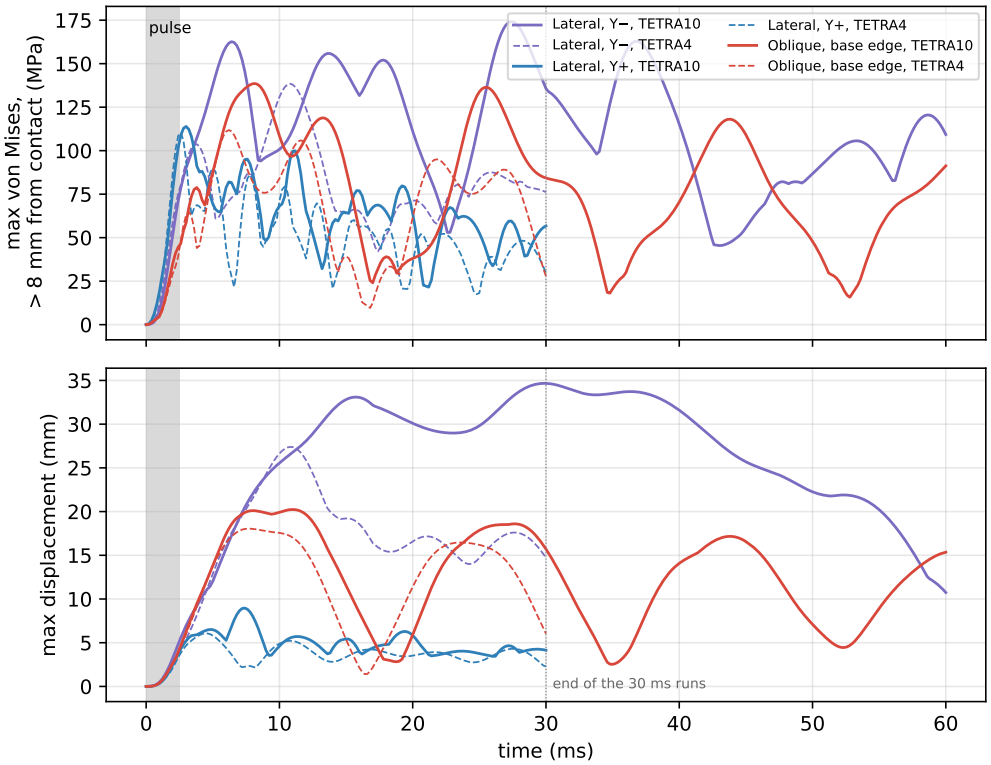


Figure 4: Impact histories at 5 % Rayleigh damping: maximum von Mises stress more than 8 mm from the contact plane (top) and maximum displacement (bottom), with quadratic (solid) and linear (dashed) elements. Grey band: the 2.5 ms pulse. Two cases with quadratic elements are continued to 60 ms.

idealisation is too severe: the patch stays welded to the floor after the pulse instead of releasing, the pulse is a uniform 251 g on the whole housing, and the material stays linear elastic. A linear model does not predict cracking; it shows that a fall of this severity is not survivable by design margin alone. This supports the direction of the June recommendation, extended to the hot spots found here: a soft bumper on the shroud and on the edge of the base, which lowers the peak deceleration by lengthening the stop, and larger fillets at the arm and stand junctions.

7 Discussion and limitations

Two results of the June analysis do not hold. Its frequencies, from linear tetrahedra, were 12 to 20 % too high for the first three modes, so the closest call it reported (mode 3 near the fourth harmonic) is a different mode near the third harmonic once the elements are converged. Its impact stresses, read at 2.5 ms, stopped before the peak of the $-y$ case and used elements that are too stiff in bending. Its oblique case fixed a slice cut through the housing instead of a contact patch. What holds is that the stress concentrates locally, at a few junctions of the housing.

The main uncertainty is the material. With E between 3.8 and 6.0 GPa, three modes cross a rotor line at 2 000 rpm; with an orthotropic printed material the mode shapes would also change, which the $\sqrt{E}$ scaling does not capture. The other limits are:

- *Material model.* Isotropic, linear elastic, no measured property; the printed part is anisotropic and weaker between layers [1, 5].

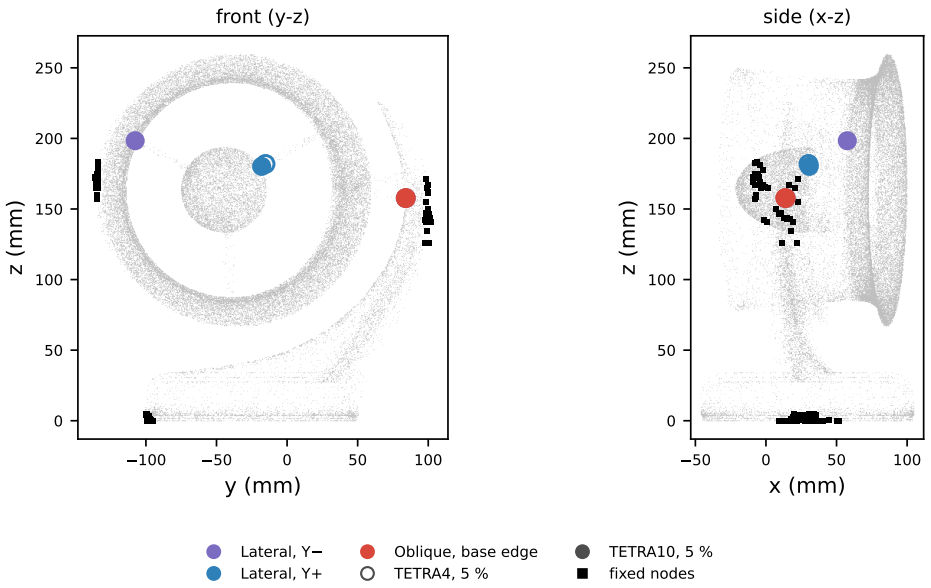


Figure 5: Fixed nodes of the three impact cases (black squares) and peak elements at 5 % damping, with linear (open circles) and quadratic (filled circles) elements. The outline is a random sample of the mesh nodes. The linear $-y$ and both oblique peaks are at the same place, where the stand joins the shroud; the two $+y$ peaks are at the junction of the upper arm with the hub.

- *Boundary conditions.* Free-free for the modes; the stand, its support and any mounting are not modelled.
- *Masses.* Motor and electronics are estimated masses spread over spherical zones, with no added stiffness; the blades are not modelled.
- *Impact.* No contact law, friction or rebound: a patch of nodes stays fixed for the whole run. The pulse (251 g, 2.5 ms, $\Delta v = 3.11$ m/s) is uniform over the housing. Damping is Rayleigh with assumed ratios. The $-y$ case reaches displacements of 30 to 40 mm, at the edge of the small-displacement assumption.
- *Validation.* No tap test, accelerometer measurement or drop test.

8 Conclusion

The free-free modes of the PETG-CF housing, converged with quadratic elements, are 44.0, 64.3 and 109.9 Hz for the first three at $E = 6$ GPa, 12 to 20 % below the values of the original linear-element model. Carried over the plausible modulus range of 3.8 GPa to 6.0 GPa, modes 3, 4 and 5 each cross a rotor line at 2 000 rpm, so the absence of resonance cannot be claimed without measuring the material. Under an idealised impact with the contact patch held fixed, the damped stress peaks at the arm and stand junctions (114 to 174 MPa with quadratic elements at 5 % damping) exceed the strength reported for printed PETG-CF. In order of priority, the next steps are: a tap test of a printed housing to measure the first frequencies and infer the effective modulus; tensile tests of printed coupons in the layer and across layers; a supported modal model with the stand on its real support; and a drop test of a printed housing with and without a bumper to calibrate the impact model.

Reproducibility statement

Inputs and scripts are in the repository: the oriented CAD (`cad/`), the meshing script (`src/mesh/mesh_carter_netgen.py`; SALOME 9.13, NETGEN, min 1 mm, max 10 mm, fineness 2), the reference modal input (`src/code_aster/`; code_aster 16.9), the convergence inputs and outputs (`data/convergence/`; code_aster 17.4), the June impact inputs (`shock/code_aster/`; code_aster 17.4) and the October impact generators, inputs, logs and time histories (`shock/extended_2026-10/`, `shock/results_extended/`; code_aster 17.4). All October runs used 2 threads on a 4-core Intel i7-6700HQ computer with 32 GB of memory: a TETRA4 modal run takes 3 to 5 min, a TETRA10 modal run about 25 min and 11 GB, a 30 ms impact run 15 to 65 min (TETRA4) or 50 to 80 min (TETRA10), and a 60 ms TETRA10 run about 100 min with stresses from 26 ms on (with stresses at every archived instant, its scratch space exceeded the 118 GB free). Meshes and result files (4 GB to 6.5 GB per impact run) are not versioned. Tables and figures are regenerated by the scripts in `paper/scripts/`, and the paper by `paper/build_paper.sh`.

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